



Cambridge (CIE) IGCSE Computer Science



Your notes

File Handling

Contents

* File Handling



Your notes

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What is file handling?

- File handling is **the use of programming techniques** to work with information stored in **text files**
- Examples of file handling techniques are:
 - opening** text files
 - reading** text files
 - writing** text files
 - closing** text files

Concept	OCR exam reference	Python
Open	OPENFILE "fruit.txt" FOR READ	file = open("fruit.txt", "r")
Close	CLOSEFILE "fruit.txt"	file.close()
Read line	READFILE "fruit.txt", LineOfText	file.readline()
Write line	OPENFILE "fruit.txt" FOR WRITE WRITEFILE "fruit.txt", "Oranges"	file.write("Oranges")
Append a file	OPENFILE "fruit.txt" FOR APPEND	file = open("shopping.txt", "a")

Pseudocode example (reading data)

Employees	Text file
<pre> OPENFILE "employees.txt" FOR READ endOfFile ← FALSE // Set end of file flag to false WHILE NOT endOfFile DO READFILE "employees.txt", name // Read line 1 (name) </pre>	<pre> Greg Sales 39000 43 Lucy Human resources </pre>



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name ← TRIM(name) // Remove extra spaces/newlines	26750
READFILE "employees.txt", department // Read line 2 (department)	28
department ← TRIM(department) // Remove extra spaces/newlines	Jordan
READFILE "employees.txt", salary // Read line 3 (salary)	Payroll
salary ← TRIM(salary) // Remove extra spaces/newlines	45000
READFILE "employees.txt", age // Read line 4 (age)	31
age ← TRIM(age) // Remove extra spaces/newlines	
IF name = "" THEN // If name is empty (end of file)	
endOfFile ← TRUE // Set end of file flag to true	
ELSE	
OUTPUT "Name: " & name // Print name	
OUTPUT "Department: " & department // Print department	
OUTPUT "Salary: " & salary // Print salary	
OUTPUT "Age: " & age // Print age	
OUTPUT "" // Add a blank line for readability	
ENDIF	
ENDWHILE	
CLOSEFILE "employees.txt"	

Python example (reading data)

Employees	Text file
<pre># Open file in read mode file = open("employees.txt", "r") endOfFile = False # Set end of file flag to false while not endOfFile: # While not end of file name = file.readline().strip() # Read line 1 and remove extra spaces/newlines department = file.readline().strip() # Read line 2 salary = file.readline().strip() # Read line 3 age = file.readline().strip() # Read line 4 if name == "": # If name is empty (end of file) endOfFile = True # Set end of file flag to true else: print("Name: ", name) # Print name print("Department: ", department) # Print department print("Salary: ", salary) # Print salary (fixed typo) print("Age: ", age) # Print age print() # Add a blank line for readability</pre>	<pre>Greg Sales 39000 43 Lucy Human resources 26750 28 Jordan Payroll 45000 31</pre>

```
# Close the file
file.close()
```



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Pseudocode example (writing new data)

Employees	Text file
<pre> OPENFILE "employees.txt" FOR APPEND // Write employee details to the file WRITEFILE "employees.txt", "Polly" // Name WRITEFILE "employees.txt", NEWLINE WRITEFILE "employees.txt", "Sales" // Department WRITEFILE "employees.txt", NEWLINE WRITEFILE "employees.txt", "26000" // Salary WRITEFILE "employees.txt", NEWLINE WRITEFILE "employees.txt", "32" // Age WRITEFILE "employees.txt", NEWLINE CLOSEFILE "employees.txt" </pre>	<pre> Greg Sales 39000 43 Lucy Human resources 26750 28 Jordan Payroll 45000 31 Polly Sales 26000 32 </pre>

Python example (writing new data)

Employees	Text file
<pre> # Open file in append mode file = open("employees.txt", "a") # Write employee details to the file file.write("Polly\n") # Name file.write("Sales\n") # Department file.write("26000\n") # Salary file.write("32\n") # Age # Close the file file.close() </pre>	<pre> Greg Sales 39000 43 Lucy Human resources 26750 28 Jordan Payroll 45000 31 Polly </pre>



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Sales
26000
32



Examiner Tips and Tricks

When opening files it is really important to make sure you use the correct letter in the open command

- "r" is for reading from a file only
- "w" is for writing to a new file, if the file does not exist it will be created. If a file with the same name exists the contents will be overwritten
- "a" is for writing to the end of an existing file only

Always make a backup of text files you are working with, one mistake and you can lose the contents!



Worked Example

Use pseudocode to write an algorithm that does the following :

- Inputs the title and year of a book from the user.
- Permanently stores the book title and year to the existing text file *books.txt* [4]

How to answer this question

- Write two input statements (title and year of book)
- Open the file
- Write inputs to file
- Close the file

Example answer

```
title = input("Enter title")
year = input("Enter year")
file = open("books.txt")
file.writeline(title)
file.writeline(year)
file.close()
```

Guidance

```
title = input("Enter title")
```

1 mark



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<code>year = input("Enter year")</code>	
<code>file = open("books.txt")</code>	1 mark
<code>file.writeline(title)</code>	1 mark
<code>file.writeline(year)</code>	
<code>file.close()</code>	1 mark